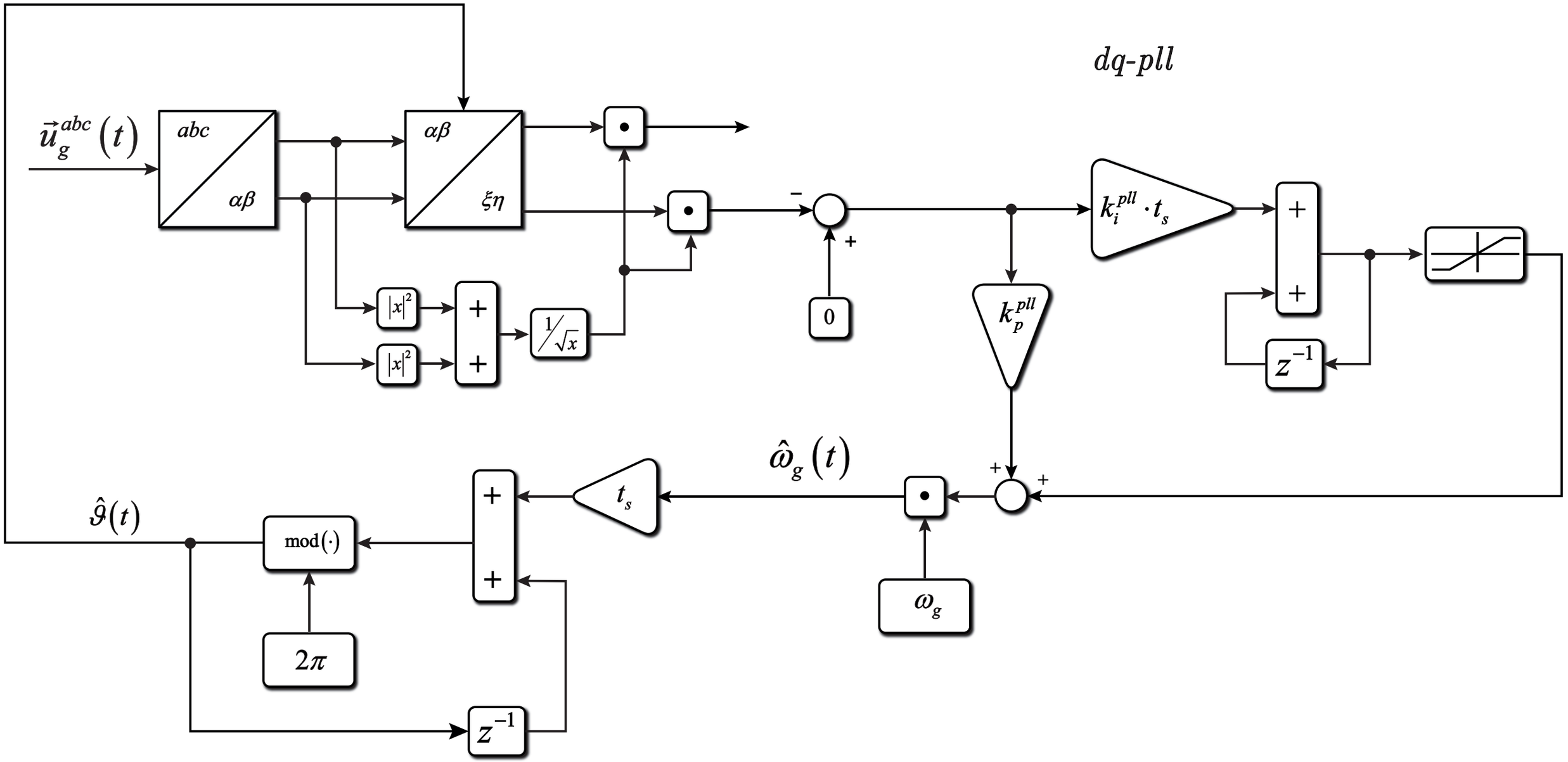

dq-pll
$\vec{u}_g^{abc}(t)$
abc
αβ
αβ
ξη
$|x|^2$
$|x|^2$
$1/\sqrt{x}$
0
k_p^{pll}
$k_i^{pll} \cdot t_s$
z^{-1}
$\hat{\omega}_g(t)$
t_s
ω_g
$\hat{\vartheta}(t)$
mod(·)
2π
z^{-1}